

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	D	A	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C,D	A,C,D	A,B,C	A,D	D	A,B,C	C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	7.00	2.40	44.84	13.86	4.00	3.00		

PART-2 : CHEMISTRY

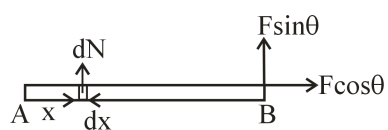
SECTION-I (i)	Q.	1	2	3	4				
	A.	A	A	D	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C,D	B,C,D	A,B,C	B,D	A,B,C	A,D	A,B,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	300.00	102.00	6.00	3.00	1.00	33.33		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	A	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C,D	A,B,C	C,D	A,B,C	A,B	A,B,C,D	A	B,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	4.00	2.25	90.00	8.00	0.25	16.00		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (C)



$$\text{For sliding to start } F \cos \theta = \int_0^l \mu_0 x dN \dots (1)$$

Balancing torque about A.

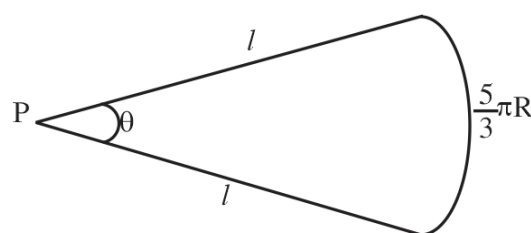
$$F \sin \theta \times l + \int_0^l dN \times x = mg \times \frac{l}{2} \dots (2)$$

From equation (1) and (2)

$$F \cos \theta = \mu_0 \left(mg \times \frac{l}{2} - F \sin \theta \times l \right)$$

$$F = \frac{\mu_0 mg \frac{l}{2}}{\cos \theta + \mu_0 l \sin \theta}; F_{\min} = \frac{\mu_0 mg l}{2\sqrt{1 + \mu_0^2 l^2}}$$

2. Ans (D)



$$\text{Electric potential at center of disc } V = \frac{\sigma R}{2\epsilon_0}$$

So potential due to given part

$$V' = \left(\frac{\sigma}{2\epsilon_0} l \right) \left(\frac{5\pi R}{3l} \right) \times \frac{1}{2\pi}$$

$$= \frac{5\sigma R}{12\epsilon_0}$$

3. Ans (A)

$$\frac{dN}{dv} = kv^2 e^{-bv^2}$$

Maxima of this function will be at most probable speed.

$$\frac{d}{dv} \left(\frac{dN}{dv} \right) = k \left[2ve^{-bv^2} + v^2 \times -2bve^{-bv^2} \right] = 0$$

$$v_{mp} = \frac{1}{\sqrt{b}}$$

$$b = \frac{1}{v_{mp}^2} = \frac{M}{2RT}$$

4. Ans (B)

$$A = A_0 e^{-\lambda t_1} \quad \dots(1)$$

$$\frac{2A}{3} = A_0 e^{-\lambda t_2} \quad \dots(2)$$

$$\frac{2}{3} = e^{\lambda(t_1 - t_2)}$$

$$\ln \left(\frac{3}{2} \right) = (t_2 - t_1) \lambda$$

$$\frac{1}{\lambda} = t_{\text{mean}} = \frac{t_2 - t_1}{\ln \left(\frac{3}{2} \right)}$$

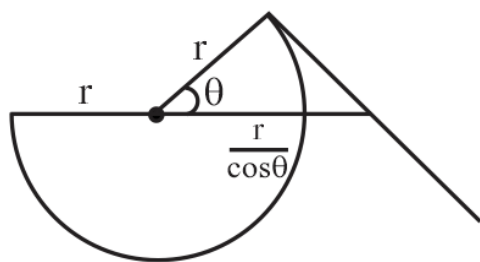
$$t_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$$

$$P(\text{survival}) = 1 - P(\text{disintegration})$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,B,C,D)



$$r + \frac{r}{\cos \theta} = d$$

$$r = \frac{d \cos \theta}{1 + \cos \theta}$$

$$\frac{2T}{r} = \rho gh$$

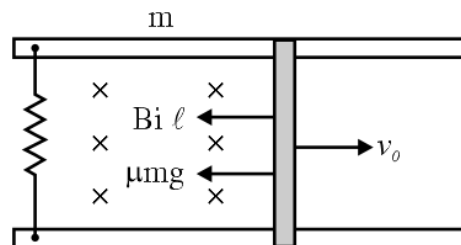
On surface, force will act due to surface tension as well. So both (C) and (D) option are INCORRECT.

6.

Ans (A,C,D)

$$B = 1 = \ell = m = R, v_0 = 2 \text{ m/s}$$

$$k = \frac{B^2 \ell^2}{mR}$$



$$-B \left(\frac{B \ell v}{R} \right) \ell - \mu mg = ma = m v \frac{dv}{dx}$$

$$\int_{v_0}^0 \frac{-v dv}{\left(\frac{B^2 \ell^2}{mR} \right) v + \mu g} = \int_0^x dx$$

$$\int \frac{v dv}{kv + \mu g} = \left[\frac{v}{k} - \frac{\mu g \ln(kv + \mu g)}{k^2} \right]_0^{v_0} = \int_0^x dx$$

$$\frac{v_0}{k} - \frac{\mu g \ln(kv_0 + \mu g)}{k^2} + \frac{\mu g \ln(\mu g)}{k^2} = x$$

$$\frac{v_0}{k} - \frac{\mu g}{k^2} \left[\ln \left(\frac{kv_0 + \mu g}{\mu g} \right) \right] = x$$

$$2 - \frac{2}{1} \ln \left(\frac{2+2}{2} \right) = x; 2 - 2 \ln 2 = x$$

$$2 - 1.39 = 0.61 \text{ meters}$$

Distance travelled before stopping.

Time taken before stopping

$$\frac{-B^2 \ell^2}{mR} v - \mu g = \frac{dv}{dt} \Rightarrow \int_{v_0}^0 \frac{-dv}{kv + b} = \int_0^t dt$$

$$\frac{\ln kv + b}{k} \Big|_0^{v_0} = t; \frac{1}{k} \ln \frac{kV_0 + b}{b} = t$$

$$\Rightarrow \ln \frac{2+2}{2} = \ln 2 = 0.693$$

$$\text{Heat in resistor} = \frac{1}{2} m v_0^2 - \mu mg x = 0.77 \text{ J}$$

7. **Ans (A,B,C)**

$V_x = u$, $a_x = 0$, as no force in x-direction,

$$V_c = v \left(1 - e^{-\frac{t}{RC}} \right)$$

$$\Rightarrow V_c = V \left(1 - \left(1 - \frac{t}{RC} + \frac{\left(\frac{t}{RC}\right)^2}{2!} - \dots \infty \right) \right)$$

$$= \frac{Vt}{RC} \quad (\because t \ll RC)$$

$$\Rightarrow E_c = \frac{V_c}{d} = \frac{Vt}{RCd}$$

$$\Rightarrow a_y = \frac{qE_c}{m} = \frac{qVt}{mdRC}$$

$$\int_0^{V_y} dV_y = \frac{qV}{mdRC} \int_0^t t dt$$

$$\Rightarrow V_y = \frac{qV}{mdRC} \frac{t^2}{2}$$

$$\Rightarrow dy = \frac{qV}{mdRC} \frac{t^2}{2} dt$$

$$\Rightarrow \frac{d}{2} = \frac{qV}{mdRC} \frac{t^3}{6}$$

$$\Rightarrow t = \left(\frac{3md^2RC}{qV} \right)^{\frac{1}{3}}$$

$$\therefore \Rightarrow x = u \times \left(\frac{3md^2RC}{qV} \right)^{\frac{1}{3}} \quad (\because x = v_x t)$$

$$V_{net} = \sqrt{V_x^2 + V_y^2}$$

$$= \sqrt{u^2 + \left(\frac{qVt^2}{2mdRC} \right)^2}$$

8. **Ans (A,D)**

$$\text{Flux through CEB} = 2 \times \frac{1}{2} \times 2 \times 2 = 4 \text{ V xm}$$

Flux through surface ABCD will be zero as

electric field in x direction and z direction will

produce same magnitude of outgoing and

incoming flux respectively.

$$\phi = \frac{Q_{en}}{\epsilon_0} \Rightarrow Q_{en} = 4\epsilon_0$$

9. **Ans (D)**

$$\text{Efficiency } \eta = 1 - \frac{Q_2}{Q_1}$$

$$Q_2 = -C_v (T_1 - T_4)$$

$$Q_1 = C_v (T_3 - T_2)$$

$$\eta = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

For adiabatic process

$$T_1 (16V_0)^{\gamma-1} = T_2 (V_0)^{\gamma-1}$$

$$T_4 (16V_0)^{\gamma-1} = T_3 (V_0)^{\gamma-1}$$

$$(T_4 - T_1) (16v)^{\gamma-1} = (T_3 - T_2) (v)^{\gamma-1}$$

$$\frac{T_4 - T_1}{T_3 - T_2} = \left(\frac{1}{16} \right)^{\gamma-1} = \frac{1}{4}$$

$$\left(\gamma_{\text{mixture}} = \frac{3}{2} \right)$$

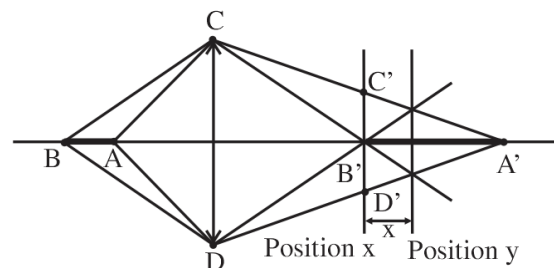
$$\eta = \frac{3}{4}$$

10. **Ans (A,B,C)**

When B is at 56, length of image

$$= \frac{50 \times 40}{10} - \frac{56 \times 40}{16} = 200 - 140 = 60 \text{ cm}$$

(10 times of object)



Radius of image when screen is 140 cm from lens.

$A'CD$ is similar to $A'C'D'$

$$C'B = \frac{4}{200} \times 60 = 1.2 \text{ mm}$$

For minimum size of image, screen should be placed at intersection of the refracted rays from A and B (position y)

$$\frac{x}{d_{\min}} = \frac{140}{4} \quad \dots (1) \quad \text{and} \quad \frac{60-x}{d_{\min}} = \frac{200}{4} \quad \dots (2)$$

Solving (1) and (2)

$$x = \frac{420}{17} \text{ cm}$$

distance from lens

$$= 140 + \frac{420}{17} = \frac{2800}{17} = 164.7 \text{ cm}$$

11. **Ans (C)**

Using dimension analysis $\Rightarrow v = k P^a r^b \ell^c \eta^d$

$$L^3 T^{-1} = k [M L^{-1} T^{-2}]^a [L]^{b+c} [M L^{-1} T^{-1}]^d$$

$$a + d = 0$$

$$-a + b + c - d = 3 \Rightarrow b + c = 3$$

$$-2a - d = -1 \Rightarrow a = 1 \Rightarrow d = -1$$

$$\text{So, } v = k \frac{P}{\eta} r^b \ell^c$$

Option which has $b + c = 3$ can be correct.

12. **Ans (A,B,C)**

$$\text{Current sensitivity} = \frac{BNA}{C}$$

$$\text{Voltage sensitivity} = \frac{BNA}{CR_g}$$

$$R_g \text{ is } \propto N$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (7.00)**

$$\vec{F} = \frac{-\partial U}{\partial x} \hat{i} - \frac{\partial U}{\partial y} \hat{j}$$

$$= -3\hat{i} - 4\hat{j}$$

$$\vec{a} = -3\hat{i} - 4\hat{j} \text{ constant acceleration}$$

It will move in a straight line of slope $\frac{4}{3}$.

$$\text{Equation of straight line } (y - 4) = \frac{4}{3}(x - 6)$$

It will intersect x axis at (3, 0) and y axis at (0, -4)

$$\text{So, } |\alpha| + |\beta| = 7$$

2. **Ans (2.40)**

$$\Delta P = B \frac{\Delta V_{\text{apparent}}}{V} = B (\gamma - 3\alpha) \Delta T$$

$$\text{Force per cm}^2 = \Delta P \times 1 \text{ cm}^2 = 2.4 \times 10^3$$

3. **Ans (44.84)**

Frequency incident on reflector

$$f' = \frac{340 + 10}{340 - 20} \times f_0$$

Frequency detected after reflection

$$f_1 = \frac{340 + 20}{340 - 20} \times f'$$

Frequency detected directly

$$f_2 = \frac{340 + 10}{340 - 10} \times f_0$$

Beat frequency = $|f_1 - f_2|$

$$= \left(\frac{35 \times 36}{32 \times 32} - \frac{35}{33} \right) \times f_0$$

$$= \frac{35(36 \times 33 - 32 \times 32)}{32 \times 32 \times 33} \times f_0$$

$$= 44.84 \text{ Hz}$$

4. **Ans (13.86)**

$ms \left(-\frac{dT}{dt} \right) = \text{Rate of heat rejection to ice} -$

Rate of which heat generated due to viscosity

Rate of heat generation = $\tau \times \omega$

$$= \omega \times 2 \int_0^R \eta \times 2\pi x dx \times \frac{\omega x}{\frac{h}{2}} \times x = \frac{2\pi\eta R^4 \omega^2}{h}$$

$$-ms \frac{dT}{dt} = \frac{T - 0}{t_0/kA} - \frac{2\pi\eta R^4 \omega^2}{h}$$

$$A = 2\pi R^2$$

$$m = \rho \times \pi R^2 h$$

$$-\int_{50}^{30} \frac{dT}{10 - T} = \frac{800}{16 \times 10^3} \int_0^t dt$$

$$t = 20 \ln \frac{10 - 50}{10 - 30}$$

$$= 20 \times 0.693 = 13.86 \text{ sec}$$

5. **Ans (4.00)**

$$C_1 = 4\pi\epsilon_0 K \times 2R$$

$$C_2 = 8\pi\epsilon_0 R$$

$$\frac{q^2}{2C_1} = \frac{1}{5} \left(\frac{q^2}{2C_1} + \frac{q^2}{2C_2} \right)$$

$$\frac{5}{C_1} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{4}{C_1} = \frac{1}{C_2} \Rightarrow K = 4$$

6. **Ans (3.00)**

By Snell's law

$$n_w \sin \theta = n_{\text{gas}} \sin 90^\circ$$

$$\theta = \frac{s}{R} = \frac{1}{3/\pi}$$

$$= \frac{\pi}{3}$$

$$\frac{4}{3} \sin 60^\circ = n_{\text{gas}} \times 1$$

$$n_{\text{gas}} = \frac{2}{\sqrt{3}}$$

$$n = 3$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A)**

Carbocation intermediate formed is most stable for IV and least for V [Back bonding].

For II and III, -I effect is considered.

2. **Ans (A)**

(A), (C) have R configuration; (B), (D) have S configuration.

3. **Ans (D)**

HNO₃ can't be used as Fe₂O₃ impurity will also dissolve in it. Fe₂O₃ + 6HNO₃ → 2Fe(NO₃)₃ + 3H₂O

4. **Ans (B)**

MnO₄⁻ shall oxidise NO₂⁻ to NO₃⁻.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,B,C,D)**

(1) [Fe₃(HCOO)₆]³⁺ → Red coloration

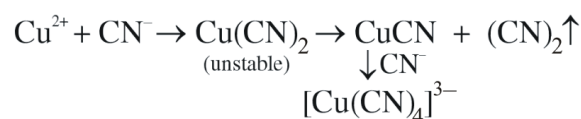
(2) [Fe₃(OH)₂(CH₃COO)₆]³⁺ → Deep red coloration

(3) (PhCOO)₃Fe.Fe(OH)₃ → Buff ppt

(4) [Fe(OPh)₆]³⁻ → Violet ppt

6. **Ans (B,C,D)**

(1)



7. **Ans (A,B,C)**

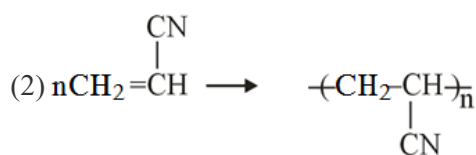
(1) TlI₃ exists as Tl⁺, I₃⁻

(2) Cr³⁺ + 3I⁻

(c) Fe²⁺ + I₂ + I⁻

8. **Ans (B,D)**

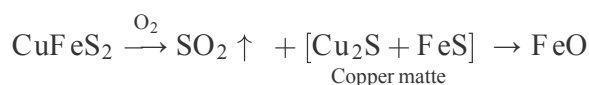
(1) Both are 2°, hence given blue color



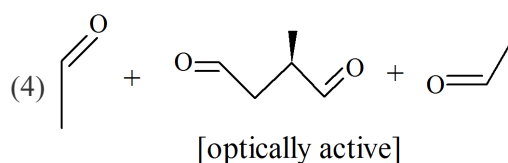
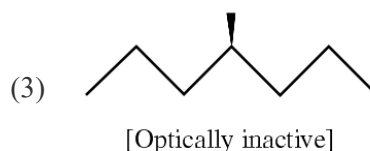
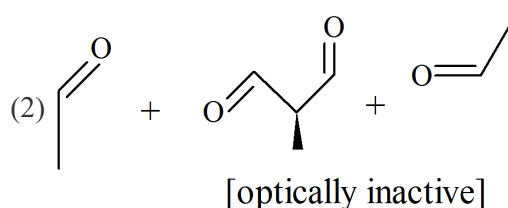
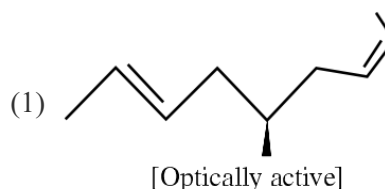
{ORLON}

(4) Hydrogen Bonding

9. **Ans (A,B,C)**



10. **Ans (A,D)**



11. Ans (A,B,C,D)

$$\text{Specific conductance, } K = \frac{1}{R} \cdot \frac{\ell}{A} = 10^{-2} \text{ S cm}^{-1}$$

∴ Equivalent conductance,

$$\begin{aligned} \Lambda^{\text{eq}} &= \frac{K \times 1000}{N} \\ &= \frac{10^{-2}}{10^{-1}} \times 10^3 \\ &= 10^2 \text{ S cm}^2 \text{ eq}^{-1} \end{aligned}$$

$$\alpha = \frac{\Lambda^{\text{eq}}}{\Lambda^{\circ}} = \frac{100}{1000} = 0.1$$

$$[H^+] = C\alpha = 0.1 \times 0.1$$

$$[H^+] = 10^{-2}$$

$$\text{pH} = 2$$

12. Ans (A,B,C)

$$\oint dS = 0; dG = V dP$$

PART-2 : CHEMISTRY

SECTION-II

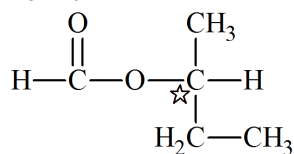
1. Ans (300.00)

$$200 = \frac{a}{3} + \frac{2b}{3} \quad \dots(i)$$

$$250 = \frac{2a}{3} + \frac{b}{3} \quad \dots(ii)$$

Solving we get a = 300

2. Ans (102.00)



3. Ans (6.00)

H-Atom	
1.	5 → 4 → 3 → 2 → 1
2.	5 → 3 → 1
3.	5 → 4 → 2 → 1
4.	5 → 4 → 1
5.	5 → 1
6.	5 → 2 → 1



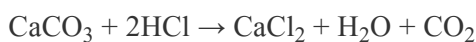
4. Ans (3.00)

PLANAR/NON- PLANAR	BOND ORDER	
↓	↓	
CO_3^{-2}	$\frac{4}{3}$	Planar
PO_4^{-3}	$\frac{5}{4}$	Non-planar
C_6H_6	$\frac{3}{2}$	Planar
SO_4^{-2}	$\frac{6}{4}$	Non-planar
$\text{CH}_3\text{COO}^{\ominus}$	$\frac{3}{2}$	Non-planar
$\text{H}_2\text{PO}_2^{\ominus}$		Non-planar
O_3	$\frac{3}{2}$	Planar

5. Ans (1.00)

Species	Orbitals used for overlapping by CA	Orbitals used for Hyb. by CA
PCl_5	5	5
XeO_4	8	4
H_2SO_4	6	4
POCl_3	5	4
IF_5	5	6
XeO_6^{-4}	10	6
$\text{ClO}_3^{\ominus}$	5	4

6. Ans (33.33)



$$n_{\text{HCl}} = \frac{200 \times 1.2}{36.5} \times \frac{7.3}{100}$$

Mass of CaCO_3 ,

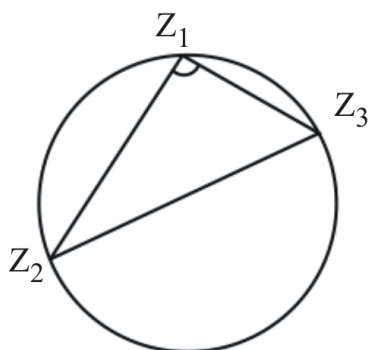
$$\begin{aligned} \text{to be taken} &= \frac{200 \times 1.2 \times 7.3}{36.5 \times 100 \times 2} \times \frac{100 \times 100}{72} \\ &= 33.33\text{g} \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A)**

$$\begin{aligned} Z_1 (\bar{Z}_2 + \bar{Z}_3) + \bar{Z}_1 (Z_2 + Z_3) &= 0 \\ \Rightarrow |Z_1|^2 + |Z_2|^2 - Z_1 \cdot \bar{Z}_2 - \bar{Z}_1 \cdot Z_2 \\ &+ |Z_1|^2 + |Z_3|^2 - Z_1 \cdot \bar{Z}_3 - \bar{Z}_1 \cdot Z_3 = 4 \\ \Rightarrow (Z_1 - Z_2) \cdot (\bar{Z}_1 - \bar{Z}_2) + (Z_1 - Z_3) \cdot (\bar{Z}_1 - \bar{Z}_3) &= 4 \\ \Rightarrow |Z_1 - Z_2|^2 + |Z_1 - Z_3|^2 &= 4 = 2^2 \end{aligned}$$

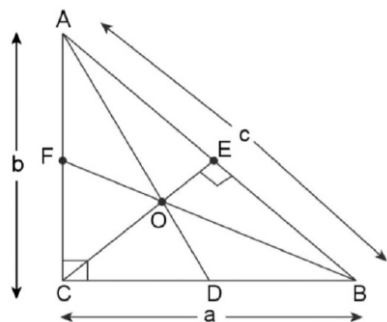


Hence, Z_2, Z_3 are diametrically endpoints

$$\begin{aligned} \Rightarrow Z_2 + Z_3 &= 0 \\ \Rightarrow |Z_1 + Z_2 + Z_3| &= 1 \end{aligned}$$

2. **Ans (C)**

Solution: Let a, b, c , denote the base, perpendicular and hypotenuse of right angled $\triangle ABC$ as shown below;



$$S_a = \sqrt{b^2 + \frac{a^2}{4}} = \sqrt{\frac{4b^2 + a^2}{4}}; S_b = \sqrt{\frac{4a^2 + b^2}{4}}$$

$$\text{and } S_c = \frac{c}{2}, c^2 = a^2 + b^2$$

$$\Rightarrow \frac{c^2}{4} = \frac{a^2 + b^2}{4} \Rightarrow \frac{c}{2} \sqrt{\frac{a^2 + b^2}{4}}$$

$$\Rightarrow S_c = \sqrt{\frac{a^2 + b^2}{4}}$$

$$\text{Now, } \Rightarrow \frac{S_a + S_b}{S_c} = \frac{\sqrt{\frac{4b^2 + a^2}{4}} + \sqrt{\frac{4a^2 + b^2}{4}}}{\sqrt{\frac{a^2 + b^2}{4}}}$$

$$\text{let } \sqrt{\frac{4b^2 + a^2}{4}} = \alpha, \sqrt{\frac{4a^2 + b^2}{4}} = \beta$$

$$\text{and } \sqrt{\frac{a^2 + b^2}{4}} = \gamma$$

Now using the fact that $\text{RMS} \geq \text{AM}$ we have

$$\sqrt{\frac{\alpha^2 + \beta^2}{2}} \geq \frac{\alpha + \beta}{2}$$

$$\text{or } \frac{\alpha + \beta}{2} \leq \frac{1}{2} \sqrt{\frac{5b^2 + 5a^2}{2}}$$

$$\Rightarrow \alpha + \beta \leq \sqrt{\frac{5b^2 + 5a^2}{2}}$$

$$\text{or } \frac{\alpha + \beta}{\gamma} \leq \sqrt{\frac{4(5b^2 + 5a^2)}{2(a^2 + b^2)}}$$

$$\text{or } \frac{\alpha + \beta}{\gamma} \leq \sqrt{10} \Rightarrow \left(\frac{\alpha + \beta}{\gamma} \right)^2 \leq 10$$

3. **Ans (A)**

Let the equation of hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, a \text{ is constant}$$

whose asymptotes are $\frac{x}{a} + \frac{y}{b} = 0$,

$$\frac{x}{a} - \frac{y}{b} = 0$$

Let there be any point $p(h, k)$ on the hyperbola

$$\frac{h^2}{a^2} - \frac{k^2}{b^2} = 1$$

k = distance of P from transverse axis

h = distance of P from any asymptotes

$$\text{Let it } \frac{x}{a} - \frac{y}{b} = 0$$

$$k = \frac{bh - ak}{\sqrt{a^2 + b^2}}$$

$$\text{Squaring } (a^2 + b^2) k^2 = (bh - ak)^2$$

$$b^2(h^2 - k^2) = 2abhk$$

$$(h^2 - k^2)^2 = 4a^2 h^2 \frac{k^2}{b^2}$$

$$\left(\because \frac{h^2}{a^2} - \frac{k^2}{b^2} = 1 \right)$$

$$(h^2 - k^2)^2 = 4a^2 h^2 \left(\frac{h^2}{a^2} - 1 \right)$$

$$(x^2 - y^2)^2 = 4x^2(x^2 - a^2)$$

$$k = 4$$

4. Ans (C)

$$f(x) = e + (1-x) \ln\left(\frac{x}{e}\right) + \int_1^x f(t) dt \quad x > 0,$$

$$f(1) = e$$

$$f'(x) = -\ln\left(\frac{x}{e}\right) + (1-x) \times \frac{1}{x} + f(x)$$

$$f'(x) = \frac{1}{x} - \ln x + f(x)$$

$$f'(x) - f(x) = \frac{1}{x} - \ln x$$

$$f'(x)e^{-x} - f(x)e^{-x} = \frac{1}{x}e^{-x} - \ln x e^{-x}$$

$$\frac{d}{dx}(f(x)e^{-x}) = e^{-x} \left(\left(\frac{1}{x} \right) - \ln x \right)$$

$$f(x) \cdot e^{-x} = c + e^{-x} \ln x, \quad f(1) = e, \quad c = 1$$

$$f(x) = e^x + \ln x$$

$$\Rightarrow g(x) = x \ln x$$

$$A = \left| \int_0^1 x \ln x \, dx \right| = \left| \frac{x^2}{2} \ln x - \frac{x^2}{4} \right|_0^1 = \frac{1}{4}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C,D)

$$\alpha + \beta = \tan \theta - 2$$

$$\alpha\beta = -\tan \theta - 1$$

$$\Rightarrow \alpha + \beta + \alpha\beta = -3$$

$$\Rightarrow (1 + \alpha)(1 + \beta) = -2$$

Since α, β are integers possible values can be :

$$(i) 1 + \alpha = 2, 1 + \beta = -1$$

$$\Rightarrow \alpha = 1, \beta = -2$$

$$\Rightarrow \tan \theta = \alpha + \beta + 2 = 1$$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$(ii) 1 + \alpha = -2, 1 + \beta = 1$$

$$\Rightarrow \alpha = -3, \beta = 0$$

$$\Rightarrow \tan \theta = \alpha + \beta + 2 = -1$$

$$\Rightarrow \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

Total 4 possible values of θ .

6. Ans (A,B,C)

$$PQ + Q^2P^2 = I$$

$$P^3 + Q^3 = 0$$

$$I - 0 = I$$

$$PQ + Q^2P^2 - P^3 - Q^3 = I$$

$$P(Q - P^2) - Q^2(Q - P^2) = I$$

$$(P - Q^2)(Q - P^2) = I$$

$$\Rightarrow P - Q^2 \text{ and } Q - P^2 \text{ both non singular}$$

$$\text{So, } (Q - P^2)(P - Q^2) = I$$

$$QP - Q^3 - P^3 + P^2Q^2 = I$$

$$\Rightarrow QP + P^2Q^2 = I$$

7. Ans (C,D)

$$P_1 = \frac{1}{2}, P_2 = P(HH) + P(T) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} = \frac{3}{4}$$

$$P_3 = \frac{5}{8}, P_n = \frac{1}{2}P_{n-1} + \frac{1}{2}P_{n-2}$$

8. Ans (A,B,C)

We know that the circle circumscribing the

triangle formed by any three tangents of a

parabola passes through the focus of the circle

circumscribing the triangle formed by the lines

$x = 1, y = 2$ and $2x + y = 6$ passes through $S_1,$

S_2 and S_3 so the equation of circum circle is

$$\text{centre} = \left(\frac{3}{2}, 3 \right),$$

$$2r = \sqrt{1^2 + 4}$$

$$2r = \sqrt{5}$$

$$a = \frac{3}{2}, b = 3$$

$$2a + b = 6$$

$$4r^2 = 5$$

Max area of a triangle

$$\Rightarrow \frac{\sqrt{3}}{4}(\sqrt{3}r)^2 = \frac{3\sqrt{3}}{4}r^2$$

9. Ans (A,B)

For a, b, c > 0

A.M. ≥ H.M.

$$\frac{a+b+c}{3} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{6} \Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{3}{2}$$

$$\text{Now, } \frac{\left(a + \frac{1}{b}\right)^2 + \left(b + \frac{1}{c}\right)^2 + \left(c + \frac{1}{a}\right)^2}{3}$$

$$\geq \left(\frac{a+b+c + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3}\right)^2$$

$$\geq \left(\frac{6 + \frac{3}{2}}{3}\right)^2$$

$$\left(a + \frac{1}{b}\right)^2 + \left(b + \frac{1}{c}\right)^2 + \left(c + \frac{1}{a}\right)^2 \geq \frac{75}{4}$$

10. Ans (A,B,C,D)

$$f\left(\frac{x+15}{2}\right) = f\left(\frac{3-x}{2}\right)$$

$$x \rightarrow 3-x$$

$$f\left(\frac{18-x}{2}\right) = f\left(\frac{x}{2}\right)$$

$$\Rightarrow f\left(9 - \frac{x}{2}\right) = f\left(\frac{x}{2}\right)$$

$$\Rightarrow f(x) = f(9-x)$$

$$\Rightarrow f(x) = -f(9-x)$$

$$\Rightarrow f(0) = -f(9) = 0 \dots\dots(1)$$

$$\Rightarrow f'\left(\frac{3}{2}\right) = f'\left(\frac{15}{2}\right) = 0 \dots\dots(2)$$

$$\Rightarrow f'(3) = f'(6) = 0 \dots\dots(3)$$

$$\Rightarrow f'(5) = f'(4) = 0 \dots\dots(4)$$

$$\Rightarrow f'\left(\frac{11}{2}\right) = f'\left(\frac{7}{2}\right) = 0 \dots\dots(5)$$

$$g(x) = f'(x) f''(x) + (f''(x))^2$$

$$= \frac{d}{dx} [f''(x) \cdot f'(x)]$$

So minimum number of roots of $f'(x) f''(x)$ is 19

So minimum number of roots is 18.

$$2k^2 = 18$$

11. Ans (A)

$$x^2 dy - 2xy dx = x^3 y^3 dx + x^4 y^2 dy$$

$$x^4 d\left(\frac{y}{x^2}\right) = x^3 y^2 (y dx + x dy) = x^3 y^2 d(xy)$$

$$x^2 \frac{d\left(\frac{y}{x^2}\right)}{y} = xy d(xy)$$

$$\frac{d\left(\frac{y}{x^2}\right)}{\frac{y}{x^2}} = xy d(xy)$$

12. Ans (B,C,D)

Let these normal meet in (h, k) then their feet lie on the curve

$$A^2 x (k-y) + A^2 y (x-h) = 0$$

$$\Rightarrow (A^2 - B^2) xy + B^2 kx - A^2 hy = 0$$

The locus passes through the point where the lines $ax + by - 1 = 0$ and $cx + dy - 1 = 0$. Then the given ellipse

$$\left(\frac{x^2}{A^2} + \frac{y^2}{B^2} - 1\right) + \lambda (ax + by - 1)(cx + dy - 1)$$

$$= (A^2 - B^2)xy + B^2 kx - A^2 hy$$

compare the co-efficient of x^2 , y^2 and constant terms

$$\frac{1}{A^2} - \lambda ac = 0, \frac{1}{B^2} - \lambda bd = 0, -1 - \lambda = 0$$

$$\frac{1}{A^2} + ac = 0, \frac{1}{B^2} + bd = 0$$

$$\Rightarrow A^2 ac = -1 \quad B^2 bd = -1$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (4.00)

$$|z| = 1,$$

$$= |z^4 + z^3 - 2z^2 i + z + 1|$$

$$= \left| z^2 \left(\left(z^2 + \frac{1}{z^2} \right) + \left(z + \frac{1}{z} \right) - 2i \right) \right|$$

$$= |z^2 + \bar{z}^2 + z + \bar{z} - 2i|, z = x + iy$$

$$= |2(x^2 - y^2) + 2x - 2i|, x^2 + y^2 = 1$$

$$= 2|2x^2 - 1 + x - i|$$

$$= 2\sqrt{(2x^2 + x - 1)^2 + 1}$$

Hence $-1 \leq x \leq 1$ and max value of $2x^2 + x - 1$ is 2

$$\text{So } |z^4 + z^3 - 2z^2 i + z + 1| \leq 2\sqrt{5}$$

2. Ans (2.25)

$$f(x) = 2x^3 - 3x^2 - x + \frac{3}{2}$$

$$f(1-x) = 2(1-x)^3 - 3(1-x)^2 - (1-x) + \frac{3}{2}$$

$$= 2(1-x^3 - 3x + 3x^2) - 3(1+x^2-2x) - 1 + x + \frac{3}{2}$$

$$= 2-2x^3-6x+6x^2-3-3x^2+6x-1+x+\frac{3}{2}$$

$$= -2x^3 + 3x^2 + x - \frac{3}{2} + 1$$

$$f(1-x) = -f(x) + 1$$

$$f(x) + f(1-x) = 1$$

$$I = \int_{1/8}^{7/8} f(f(x)) \cdot dx = \int_{1/8}^{7/8} f[1-f(x)] dx$$

$$I = \int_{1/8}^{7/8} (1-f(f(x))) dx$$

$$2I = \int_{1/8}^{7/8} 1 dx = \frac{3}{4} \Rightarrow I = \frac{3}{8}$$

$$6I = 6 \times \frac{3}{8} = \frac{18}{8} = 2.25$$

3. Ans (90.00)

5 digit number is abcde

$$= 10000a + 1000b + 100c + 10d + e = n$$

4 digit number will be (deleting middle digit)

$$1000a + 100b + 10d + e = m$$

$\Rightarrow \frac{n}{m}$ is an integer

$$\frac{10000a + 1000b + 100c + 10d + e}{1000a + 100b + 10d + e} = 10$$

$$100C = 90d + 9e$$

where c, d, e $\in \{0, 1, 2, 3, \dots, 9\}$

'e' must be 0

$$100c = 90d \Rightarrow 10c = 9d$$

c and d should be integers from 0 to 9.

$$c = d = 0$$

$$c = d = e = 0$$

$$\text{Total ways} = 9 \times 10 = 90$$

4. Ans (8.00)

Opposite edges of tetrahedron are $x + z = 0$ and

$$y + z = 0 \Rightarrow \frac{x}{1} = \frac{y}{1} = \frac{z}{-1} \dots \dots \dots (1)$$

$$\text{and } x + y = 0 \text{ and } x + y + z = 4\sqrt{6}$$

$$\Rightarrow \frac{x}{1} = \frac{y}{-1} = \frac{z-4\sqrt{6}}{0} \dots \dots \dots (2)$$

If $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$ be the

equation of two lines then

$$S.D = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{\vec{b}_1 \times \vec{b}_2} \right|$$

$$\text{So here } \vec{a}_2 - \vec{a}_1 = 4\sqrt{6}\hat{k}$$

$$\vec{b}_1 = \hat{i} + \hat{j} - \hat{k}, \vec{b}_2 = \hat{i} - \hat{j}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & -1 & 0 \end{vmatrix} = -\hat{i} - \hat{j} - 2\hat{k}$$

$$\Rightarrow S.D. = \left| \frac{8\sqrt{6}}{\sqrt{6}} \right| = 8$$

5. Ans (0.25)

$$x \int_0^x f(t) dt + \int_0^x (1-t)f(t) dt = x^4 + x^2$$

Differentiate with respect to x

$$x \cdot f(x) + \int_0^x f(t) dt + (1-x)f(x) = 4x^3 + 2x$$

Again differentiate with respect to x

$$f(x) + f(x) = 12x^2 + 2$$

$$\frac{12}{\pi} \int_0^1 \frac{dx}{12(x^2 + 1)} = \frac{12}{12 \times \pi} \tan^{-1} x \Big|_0^1 = \frac{1}{4}$$

6. **Ans (16.00)**

$$\frac{(1+x)^{20} + (1-x)^{20}}{2}$$

$$= {}^{20}C_0 + {}^{20}C_2 \cdot x^2 + {}^{20}C_4 \cdot x^4 + \dots + {}^{20}C_{20} x^{20}$$

Also,

$$\frac{(x+1)^{20} + (x-1)^{20}}{2} = {}^{20}C_0 x^{20} + {}^{20}C_2 \cdot x^{18} + \dots + {}^{20}C_{20}$$

Now, $\sum {}^{20}C_{2r} \cdot {}^{20}C_{2r+2} = \text{coefficient of } x^{22}$

$$\text{in } \frac{\left((1+x)^{20} + (1-x)^{20} \right)^2}{4} =$$

$$\frac{(1+x)^{40} + (1-x)^{40} + 2(1-x^2)^{20}}{4}$$

$$= \frac{1}{4} \left[{}^{40}C_{22} + {}^{40}C_{22} - 2 {}^{20}C_{11} \right]$$

$$= \frac{1}{2} \left[{}^{40}C_{22} - {}^{20}C_{11} \right]$$

$$= \frac{1}{2} \left[\frac{40}{22} \cdot {}^{39}C_{21} - \frac{20}{11} \cdot {}^{19}C_{10} \right]$$

$$= \frac{10}{11} \left[{}^{39}C_{21} - {}^{19}C_{10} \right]$$

$$\Rightarrow a = 10$$

$$xy = 40 = 2^3 \cdot 5^1$$

will have $2 \times 2 \times 4 = 16$

ordered pairs



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

08-07-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,B,C	A,B,C,D	A,B,D	A	A	A,C	A,B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	A	A	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	1.50	1.25	2.73	0.44	48.00 to 49.00	50.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	D	A,B,C	A,C,D	A	B,C,D	A,B,D	A,B,D	C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	B	D	B	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	8.00	9.00	4.00	6.00	1.50	6.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	B,D	A,C,D	A,B	B,C	A,D	B,C	B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	D	C	B	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	1.60	2.82	15.00	75.00	2.25	23.55		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C)

By work energy theorem

$$WD = 4kE$$

$$W_{ex} + W_g + W_f = 0$$

(1) While going up

$$W - mgh - (\mu mg \cos \theta) L = 0.$$

$$W_1 = mgh + \mu mg (L \cos \theta) \quad \{(L \cos \theta = x)\}$$

(2) While slide down

$$W + mgh - \mu mg \cos \theta L = 0$$

$$W_2 = \mu mgx - mgh$$

$$\text{Total work } W_1 + W_2 = 2\mu mgx$$

$$\frac{W}{2\mu mg} = x$$

Also it's given that block doesn't slide on plane

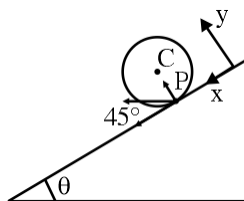
$$\mu \geq \tan \theta; \frac{W}{2mgx} \geq \tan \theta$$

$$\text{Also } \frac{h}{x} = \tan \theta \Rightarrow x = \frac{h}{\tan \theta}$$

$$\frac{W \tan \theta}{2mgh} \geq \tan \theta; h \leq \frac{W}{2mg} = \frac{40}{40}$$

$$h \leq 1m$$

2. Ans (A,B,C)



Condition of pure rolling

$$\mu \geq \frac{1}{3} \tan \theta$$

$$\mu \geq \frac{1}{3} \times \frac{1}{4}$$

$$\mu \geq \frac{1}{4}$$

Given $\mu = \frac{1}{5} = 0.2$ so cylinder is slipping

acceleration of center c

$$a_c = g \sin \theta - \mu g \cos \theta = \frac{22}{5} \text{ m/s}^2$$

$$\alpha R = \frac{\mu m g \cos \theta R}{\frac{1}{2} m R^2} = 2 \mu g \cos \theta$$

$$= 2 \times \frac{1}{5} \times 8 = \frac{16}{5} \text{ m/s}^2$$

$$a_{px} = a_c - \alpha R = \frac{6}{5} \text{ m/s}^2$$

$$\omega = \alpha t = \frac{16}{5R} t$$

$$a_{py} = \omega^2 R = \frac{256 t^2}{5R},$$

As per condition in question.

$$a_{py} = a_{px} \rightarrow \frac{256}{25R} t^2 = \frac{6}{5} \rightarrow t^2 = \frac{30R}{256}$$

$$t^2 = \frac{30}{256} \times \frac{1}{30} = \frac{1}{256}$$

$$\Rightarrow t = \frac{1}{16} \text{ sec}$$

Work done by friction in ground frame on the Cylinder in is

$$= -\mu m g \cos \theta \times S_p = -\mu m g \cos \theta \times \frac{1}{2} a_{px} t^2$$

$$= -\frac{1}{5} \times 1 \times 10 \times \frac{4}{5} \times \frac{1}{2} \times \frac{6}{5} \times 1$$

$$= -\frac{240}{250} = -0.96$$

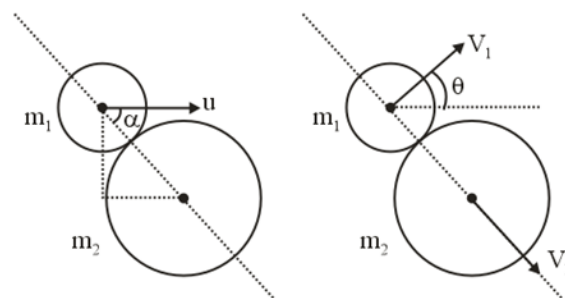
Work done by friction in the C frame on the cylinder in $\frac{1}{16}$ sec. equals the change in K.E. in C frame of the cylinder.

$$\omega = \frac{16t}{5R} = \frac{16}{5} = \frac{1}{\left(\frac{1}{30}\right)} \times \frac{1}{16} = 6 \text{ rad/s}$$

$$W = \Delta K = \frac{1}{2} \left(\frac{1}{2} \times 1 \times \left(\frac{1}{30} \right)^2 \right) = 36 = \frac{9}{900} = \frac{1}{100} \text{ J} = 0.01 \text{ J}$$

Work done by friction overall is independent of reference frame.

3. Ans (A,B,C,D)



$$V_C = \frac{m_1 u}{m_1 + m_2}$$

In c frame speed before collision equals speed after collision. Speed before collision

$$= u - v_c = \frac{m_2 u}{m_1 + m_2}$$

$$u \sin \alpha = v_1 \sin(\theta + \alpha)$$

$$v_2 - u \cos \alpha = v_1 \cos(\theta + \alpha)$$

$$m_1 u \cos \alpha = m_1 v_1 \cos(\theta + \alpha) + m_2 v_2$$

Taking $\frac{v_1}{u}$, $\frac{v_2}{u}$ & $\frac{m_2}{m_1}$ as r_1 , r_2 & μ . These equations become.

$$\sin \alpha = r_1 \sin(\theta + \alpha)$$

$$r_2 - \cos \alpha = r_1 \cos(\theta + \alpha)$$

$$\mu r_2 + r_1 \cos(\theta + \alpha) = \cos \alpha$$

Solving these equations we get

$$r_1 \cos(\theta + \alpha) [1 + \mu] = (1 - \mu) \cos \alpha$$

$$r_1 \cos(\theta + \alpha) = \left(\frac{1 - \mu}{1 + \mu} \right) \cos \alpha$$

$$r_1 \cos(\theta + \alpha) = M \cos \alpha$$

$$\left(\text{where } M = \frac{1 - \mu}{1 + \mu} \right)$$

$$\text{Also } r_1 \sin(\theta + \alpha) = \sin \alpha$$

Solving them we get

$$\tan \theta = \frac{\tan \alpha (1 - M)}{\tan^2 \alpha + M}$$

For maximum value of θ we will differentiate

$\tan \theta$ wrt α and put the value equal to zero

$$0 = (\tan^2 \alpha + M)(1 - M) \sec^2 \alpha$$

$$-(2 \tan \alpha \sec^2 \alpha)(1 - M) \tan \alpha$$

$$\Rightarrow \tan \alpha = \sqrt{M}$$

Substituting $\tan \alpha$ in $\tan \theta$ equation we get

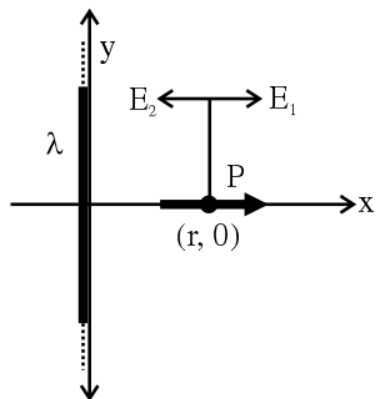
$$\sin \theta = \frac{m_2}{m_1}$$

4. Ans (A,B,D)

$$(A) \vec{E}_1 + \vec{E}_2 = 0$$

$$|\vec{E}_1| = |\vec{E}_2| \Rightarrow \frac{2k\lambda}{r} = \frac{kp}{y^3}$$

$$y = \left(\frac{pr}{2\lambda} \right)^{\frac{1}{3}}$$

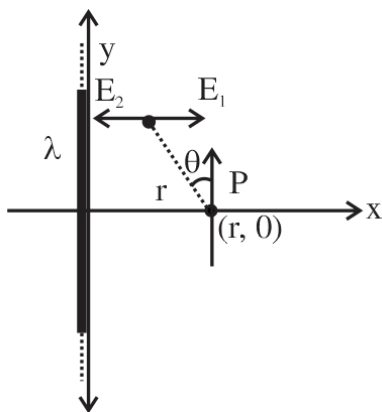


$$(B) \vec{E}_1 + \vec{E}_2 = 0$$

Possible only for $\tan \theta = \sqrt{2}$

$$\frac{kp}{r^3} \sqrt{1 + 3\cos^2 \theta} = \frac{2k\lambda}{r(1 - \sin \theta)}$$

$$\frac{p}{\lambda} = \sqrt{2} (\sqrt{3} + \sqrt{2}) r^2$$



$$(C) U = -\vec{P} \cdot \vec{E}$$

$$F = -\frac{dU}{dr} = \vec{P} \cdot \frac{d\vec{E}}{dr} = P \cdot \frac{d}{dr} \left(\frac{2k\lambda}{r} \right)$$

$$F = p2k\lambda \frac{dr^{-1}}{dr} = \frac{-2kp\lambda}{r^2}$$

$$\vec{F} = -\frac{\lambda P}{2\pi\epsilon_0 r^2} \hat{i}$$

$$(D) F = p \frac{dE}{dr} \cos 90^\circ = 0$$

5. Ans (A)

For liquid to not come out

$$\Delta v_{\text{liquid}} \leq \Delta v_{\text{container}}$$

$$v_0 \gamma \Delta T \leq v_0 3\alpha \Delta T$$

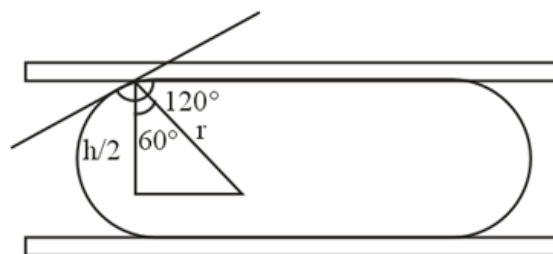
$$\gamma \Delta T \leq 3\alpha \Delta T$$

$$\gamma \leq 3\alpha \quad \text{if } \Delta T = +ve$$

$$\gamma \geq 3\alpha \quad \text{if } \Delta T = -ve$$

Common condition is $\gamma = 3\alpha$.

6. Ans (A)



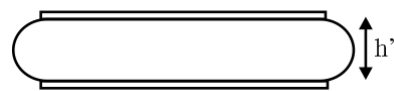
$$\frac{r}{h/2} = \cos 60^\circ \Rightarrow r = h$$

$$\text{Pressure excess in liquid} = \frac{T}{h}$$

On balancing forces on upper plate

$$mg = \frac{T}{h} \times \pi \left(\frac{R}{2} \right)^2$$

When extra mass is put and liquid is about to extrude, angle of contact will become 180° .



$$h \times \frac{\pi R^2}{4} = \pi R^2 \times h' \Rightarrow h' = \frac{h}{4}$$

$$\text{Pressure excess inside} = \frac{T}{h'/2} = \frac{T}{h/8}$$

$$mg + \Delta mg = 8 \times \frac{T}{h} \times \pi R^2$$

$$\Delta m = 32m - m = 31m = 31 \times 18$$

$$\Delta m = 558 \text{ g}$$

7. Ans (A,C)

Let three waves are

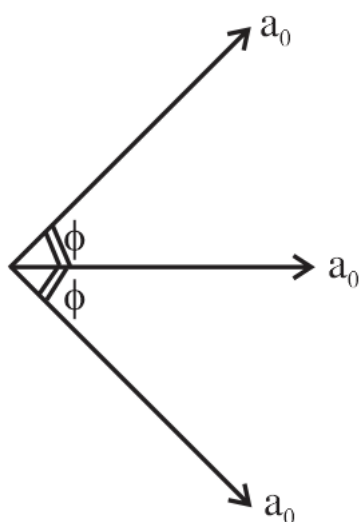
$$y_1 = a_0 \sin(\omega t - kx - \phi)$$

$$y_2 = a_0 \sin(\omega t - kx)$$

$$y_3 = a_0 \sin(\omega t - kx + \phi)$$

$$\text{Phase} = \frac{2\pi}{\lambda} \Delta x = \phi$$

By phasor method



$$\Rightarrow a_{\text{net}} = a_0 + 2a_0 \cos \phi.$$

$$A = a_0 [1 + 2 \cos \phi]$$

$$= a_0 \left[1 + 2 \cos \left(\frac{2\pi}{\lambda} \Delta x \right) \right]$$

$$A = 0 = 1 + 2 \cos \left(\frac{2\pi}{\lambda} \Delta x \right)$$

$$\frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{3} \quad \text{OR} \quad \frac{2\pi}{\lambda} \Delta x = \frac{4\pi}{3}$$

$$\Delta x_1 = \frac{\lambda}{3} \quad \Delta x_2 = \frac{2\lambda}{3}$$

$$A = A_{\text{max}} \Rightarrow 3a_0$$

$$\frac{2\pi}{\lambda} \Delta x = 2n\pi$$

$$\Delta x = n\lambda$$

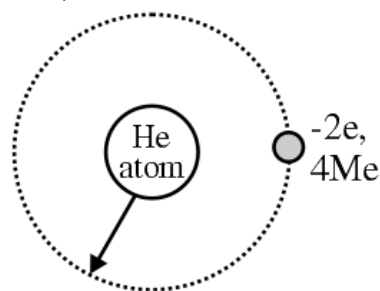
$$\text{At } \Delta x = \frac{\lambda}{2}$$

$$A = a_0$$

$$I_1 = k(9a_0^2) \quad I_2 = K a_0^2$$

$$\frac{I_1 + I_2}{I_1 - I_2} = \frac{10ka_0^2}{8ka_0^2} = \frac{5}{4}$$

8. Ans (A,B,C)



$$\frac{mV^2}{r} = \frac{(2e)q}{4\pi\epsilon_0 r^2}$$

$$mVr = \frac{nh}{2\pi}$$

$$r = \frac{nh}{2\pi mV}$$

$$r = \frac{\epsilon_0 h^2}{8\pi e^2 m} \times \frac{n^2}{Z}$$

$$\text{For } n = 1 \text{ and } Z = 2$$

$$r = 0.033 \text{ \AA}$$

$$\text{Similarly } V = \frac{2e^2}{\epsilon_0 h}$$

$$\text{Energy of } n^{\text{th}} \text{ state } E_n = -\frac{2me^4}{\epsilon_0^2 h^2} \times \frac{Z^2}{n^2}$$

$$\text{Energy of series limit of } 1^{\text{st}} \text{ excited state will be } = 13.6 \times 16 = 217.6 \text{ eV}$$

PART-1 : PHYSICS

SECTION-I (ii)

9. Ans (C)

$$y = A \sin kx \sin \omega t$$

Instantaneous power

$$P = -\frac{\omega k A^2 T}{4} \sin 2kx \sin 2\omega t$$

$$\text{At } t = \frac{1}{8f_0}, P = -\frac{\omega k A^2 T}{4} \sin 2 \times \frac{2\pi}{\lambda} x$$

$$\text{at } x = \frac{\ell}{2}, P = 0$$

$$\text{at } x = \frac{\ell}{3}, P = +\frac{\omega k A^2 T}{8}$$

$$\text{at } x = \frac{\ell}{4}, P = 0$$

$$\text{at } x = \frac{\ell}{6}, P = -\frac{\omega k A^2 T}{8}$$

10. **Ans (A)**

$$\text{At } t = \frac{1}{4f_0}, \sin 2\omega t = 0$$

So instantaneous power will be zero everywhere

11. **Ans (A)**

$$WD = \text{Area under the graph} = P_0 V_0 = \frac{RT_0}{3}$$

$$\Delta U = \frac{f}{2} nR\Delta T = \frac{f}{2} (P_f V_f - P_i V_i)$$

$$= \frac{3}{2} \left(\frac{3P_0}{2} 2V_0 - P_0 V_0 \right)$$

$$= 3P_0 V_0 = \frac{3RT_0}{3}$$

$$Q = W + \Delta U$$

$$= \frac{RT_0}{3} + RT_0 = \frac{4}{3} RT_0$$

$$Q_{1 \rightarrow 2} = nC_p \Delta T$$

$$= \frac{5}{2} (nR\Delta T)$$

$$= \frac{5}{2} (P_2 2V_0 - P_0 V_0) = \frac{5}{2} P_0 V_0$$

$$= \frac{5}{6} RT_0$$

12. **Ans (D)**

$$WD_{2 \rightarrow 3} = 0$$

$$WD_{1 \rightarrow 2} = nR \frac{T_0}{3} \ln \left(\frac{2V_0}{V_0} \right)$$

$$\Delta U = \frac{f}{2} nR\Delta T$$

$$= \frac{3}{2} R \left(T_0 - \frac{T_0}{3} \right)$$

$$= RT_0$$

$$\Delta Q = WD + \Delta U$$

$$= RT_0 + \frac{RT_0}{3} \ln 2$$

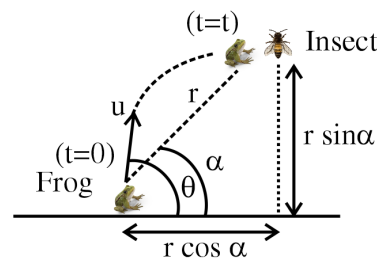
$$\Delta Q_{1 \rightarrow 2} = W_{1 \rightarrow 2}$$

$$= \frac{RT_0}{3} \ln 2$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (1.50)**



$$T = \frac{2u \sin(\theta - \alpha)}{g \cos \alpha}$$

$$\delta_x = u_x T$$

$$\Rightarrow r \cos \alpha = u \cos \theta \times \frac{2u \sin(\theta - \alpha)}{g \cos \alpha}$$

$$\Rightarrow u^2 = \frac{gr \cos^2 \alpha}{2 \sin(\theta - \alpha) \cos \theta}$$

$$\Rightarrow u^2 = \frac{gr \cos^2 \alpha}{\sin(2\theta - 2\alpha) - \sin \alpha}$$

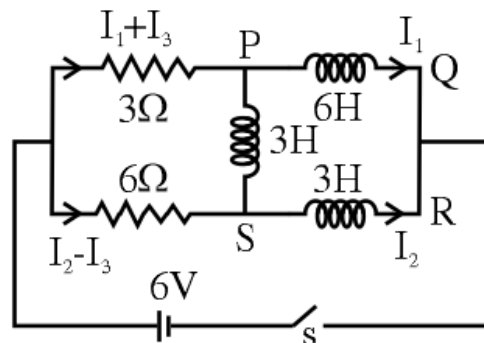
$$\text{For } u = u_{\min} \Rightarrow 2\theta - \alpha = 90^\circ$$

$$\text{and } U_{\min}^2 = \frac{gr \cos^2 \alpha}{1 - \sin \alpha} = 2r(1 + \sin \alpha)$$

$$= 10 \times 0.1 \times \left(1 + \frac{1}{2} \right)$$

$$= 1.5 \text{ m/s}$$

2. **Ans (1.25)**



In loop PQRSP

$$-\frac{6dI_1}{dt} + \frac{3dI_2}{dt} + \frac{3dI_3}{dt} = 0$$

$$\Rightarrow 2dI_1 = dI_2 + dI_3$$

$$\Rightarrow 2I_1 = I_2 + I_3 \dots (1)$$

Also at steady state

$$I_1 + I_3 = 2 \dots (2)$$

$$I_2 - I_3 = 1 \dots (3)$$

$$\text{Solving this we get } I_1 = \frac{5}{4} = 1.25 \text{ A}$$

4. Ans (0.44)

The smallest width that may be resolved is
proportional to wavelength

Electron microscope $d_1 = K\lambda_e$

$$\Rightarrow \lambda_1 = \frac{h}{\sqrt{2m_e(K.E)}}$$

Light microscope $d_2 = K\lambda_p$

$$\lambda_p = \frac{hc}{(K.E)} \quad (\text{Given } E_{\text{photon}} = K.E \text{ of } e^-)$$

$$\begin{aligned} \therefore \frac{\lambda_e}{\lambda_p} &= \frac{\left(\frac{h}{\sqrt{2m_e(K.E)}} \right)}{\left(\frac{hc}{(K.E)} \right)} = \frac{(K.E)}{(\sqrt{2m_e(K.E)})} \\ &= \sqrt{\frac{(K.E)}{2m_e c^2}} \\ &= \sqrt{\frac{200 \times 10^3 (1.6 \times 10^{-19})}{2 \times (9 \times 10^{-31}) \times (3 \times 10^8)^2}} \\ &= \frac{4}{9} = 0.44 \end{aligned}$$

5. Ans (48.00 to 49.00)

$$\begin{aligned} \text{KE of daughter : } K_1 &= \frac{Q(A-4)}{A} \\ &= \frac{(1505.5 + 28.3 - 1480) \times 200}{204} = 53 \text{ MeV} \end{aligned}$$

$$v = \sqrt{\frac{2K_1}{M}} = 7.28 \times 10^6 \text{ m/s}$$

Energy of γ -photon ; $E = 2 \text{ MeV}$

$$M\Delta v = \frac{E}{C} \Rightarrow \Delta v = \frac{10}{3} \times 10^3 \text{ m/s}$$

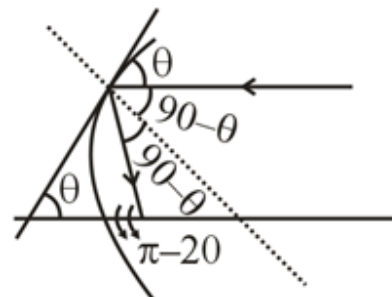
$$\frac{\Delta K_1}{K_1} = \frac{2\Delta v}{v}$$

$$\Rightarrow \Delta K_1 = \frac{2\Delta v}{V_1} K_1 = 48.53 \times 10^3 \text{ eV}$$

6. Ans (50.00)

Refraction at plane surface

$$\frac{1.5}{v} = \frac{1}{40} \quad v = 60 \text{ cm}$$



Reflection at curved surface.

$$\frac{1}{v} - \frac{1.5}{-90} = \frac{1 - 1.5}{-30}$$

$$v = \infty$$

$$\tan \theta = \frac{dx}{dy} = \frac{1}{y} \bigg|_{x_1 y_1}$$

$$\tan(\pi - 2\theta) = -\tan 2\theta$$

$$\Rightarrow -\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{-2 \left(\frac{1}{y_1} \right)}{1 - \left(\frac{1}{y_1} \right)^2}$$

$$\tan(\pi - 2\theta) = \frac{-2y_1}{y_1^2 - 1} = -\left(\frac{y_1 - 0}{x_1 - x} \right)$$

$$2x_1 - 2x = 2x_1 - 1$$

$$x = \frac{1}{2}, \quad x = 50 \text{ cm}$$

Alternative Solution:

Parallel beam passes through focus of parabola

compare with $y^2 = 4ax$

$$4a = 2, \quad a = \frac{1}{2}$$

focus = $(a, 0) = (0.5 \text{ m}, 0)$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (D)

90th electron shall occupy 5-f orbital

2. **Ans (A,B,C)**

$$k_{b1} = 8.1 \times 10^{-5} = \frac{[\text{OH}^-][\text{enH}^+]}{[\text{en}]}$$

$$k_{b2} = \frac{[\text{OH}^-][\text{enH}_2^{2+}]}{[\text{enH}^+]}$$

$$[\text{OH}^-] \simeq [\text{enH}^+]; [\text{en}] \simeq 0.09 \text{ M}$$

$$\Rightarrow [\text{OH}^-] = [\text{enH}^+] = 2.7 \times 10^{-3} \text{ M}$$

$$[\text{enH}_2^{2+}] = k_{b2} = 7.0 \times 10^{-8}$$

3. **Ans (A,C,D)**

No. of radial nodes for 5d orbitals are 2

4. **Ans (A)**

$k_p = k_c$; $\Delta n_g = 0$; Hence, mole ratio remains constant.

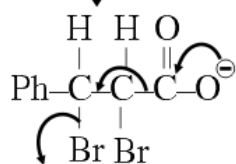
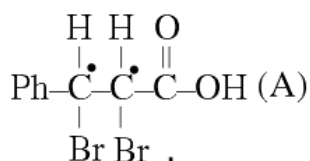
5. **Ans (B,C,D)**

Factual

6. **Ans (A,B,D)**

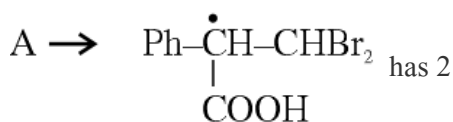
D.U of A

= 5



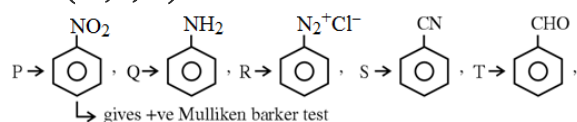
A $\rightarrow$ has 2 stereocentres. 1 mole of CO_2 is liberated from NaHCO_3 and 2nd mole from A itself.

Another possible



stereoisomers.

7. **Ans (A,B,D)**



8. **Ans (C,D)**

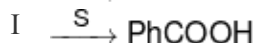
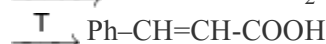
Strong ligands (from water onwards) give $\Delta_0 > P$ with Co^{3+} .

Also, 4d & 5d series metal ions have $\Delta_0 > P$ irrespective of ligand strength.

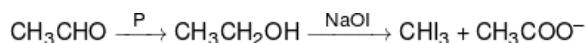
PART-2 : CHEMISTRY

SECTION-I (ii)

9. **Ans (B)**



Also,



PART-2 : CHEMISTRY

SECTION-II

1. **Ans (8.00)**

$$\lambda = \frac{RT}{\sqrt{2}\pi\sigma^2 \times P N_A}$$

$$= \frac{8.314 \times 300}{\sqrt{2} \times \pi \times (1.5 \times 10^{-10})^2 \times (\frac{1}{\pi} \times 10^{-14} \times 10^5) \times 6 \times 10^{23}}$$

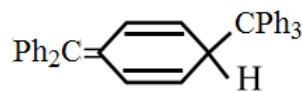
$$= \frac{8.314 \times 300 \times \sqrt{2} \times 10^6}{2 \times 2.25 \times 6}$$

$$= 130.6 \times 10^6 = 1.31 \times 10^8 \text{ m}$$

2. **Ans (9.00)**

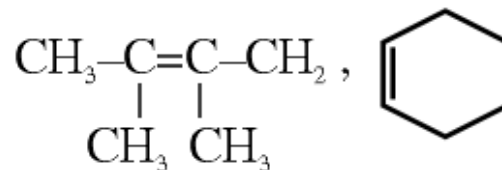
(A) Aldotetrose

(B) Kolbe's electrolysis gives $\text{Ph}_3\text{C}\cdot$



[Total 5 phenyl rings]

3. **Ans (4.00)**



4. **Ans (6.00)**

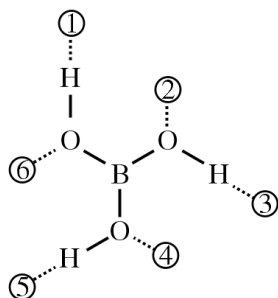
TiO_2 sol, methylene blue sol and $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ are positively charged sols.

5. Ans (1.50)

For 2nd order reaction, $t_{7/8} = 7 \times t_{1/2}$;

For 3rd order reaction $t_{3/4} = 5 \times t_{1/2}$.

6. Ans (6.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C,D)

$$f_1(x) = \frac{\pi}{2} - \sin^2 x$$

$$f_2(x) = \frac{\pi}{2} - \cos^2 x$$

$$f_3(x) = \frac{\pi}{2} - \cos^2 x$$

$$f_4(x) = \frac{\pi}{2} - \sin^2 x$$

2. Ans (B,D)

$$A = \sum_{r=1}^3 \log |\tan(60^\circ - \alpha_r)|$$

$$B = \sum_{r=1}^3 \log |\tan(60^\circ + \alpha_r)|$$

$$\Rightarrow A + B = \log |\tan(60 - \alpha_1) \tan(60 + \alpha_1)|$$

$$+ \log |\tan(60 - \alpha_2) \tan(60 + \alpha_2)|$$

$$+ \log |\tan(60 - \alpha_3) \tan(60 + \alpha_3)|$$

$$= \log \left| \frac{\tan 3\alpha_1}{\tan \alpha_1} \right| + \log \left| \frac{\tan 3\alpha_2}{\tan \alpha_2} \right| + \log \left| \frac{\tan 3\alpha_3}{\tan \alpha_3} \right|$$

$$(\because \tan(60 - \theta) \tan(60 + \theta) \tan \theta = \tan 3\theta)$$

$$= \log \left| \frac{\tan 3\alpha_1 \cdot \tan 3\alpha_2 \cdot \tan 3\alpha_3}{\tan \alpha_1 \tan \alpha_2 \tan \alpha_3} \right|$$

$$= \log \left| \frac{\tan \theta \cdot \tan \frac{\theta}{3} \cdot \tan \frac{\theta}{9}}{\tan \frac{\theta}{3} \cdot \tan \frac{\theta}{9} \cdot \tan \frac{\theta}{27}} \right| = \log \left| \frac{\tan \theta}{\tan \frac{\theta}{27}} \right|$$

$$= \log \left| \frac{\tan \left(\frac{9\pi}{4} \right)}{\tan \left(\frac{9\pi/4}{27} \right)} \right|$$

$$= \log \left| \frac{1}{\tan \frac{\pi}{12}} \right| = \log(2 + \sqrt{3}) = 1$$

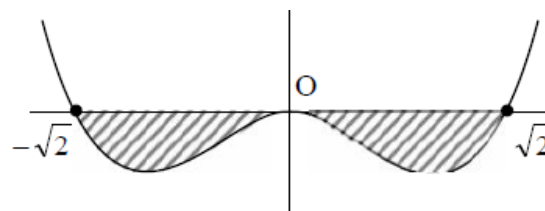
$$(\because \text{base of logarithm is } 2 + \sqrt{3})$$

3. Ans (A,C,D)

$$\text{Here } f(x) = x^4 - 2x^2 = x^2(x^2 - 2)$$

$$f'(x) = 4x(x^2 - 1)$$

Graph of $f(x)$



$$g(x) = \begin{cases} x^4 - 2x^2, & -2 \leq x < -1 \\ -1, & -1 \leq x < 0 \\ 0, & 0 \leq x < \sqrt{2} \\ x^4 - 2x^2, & \sqrt{2} \leq x \leq 2 \end{cases}$$

4. Ans (A,B)

$$\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$$

$$\sqrt{3}(\cos x + \sin x) + (\sin x - \cos x) = 2\sqrt{2} \sin 2x$$

$$\sqrt{3} \left[\frac{\cos x + \sin x}{\sqrt{2}} \right] + \left(\frac{\sin x - \cos x}{\sqrt{2}} \right) = 2 \sin 2x$$

$$\sqrt{3} \left[\sin \left(x + \frac{\pi}{4} \right) \right] - \cos \left(x + \frac{\pi}{4} \right) = 2 \sin 2x$$

$$\sin \left(x + \frac{\pi}{4} - \frac{\pi}{6} \right) = \sin 2x$$

$$2x = n\pi + (-1)^n \left(x + \frac{\pi}{12} \right)$$

$$n = 0 \Rightarrow x = \frac{\pi}{12}$$

$$n = 1 \Rightarrow 3x = \pi - \frac{\pi}{12} = \frac{11\pi}{12}$$

$$x = \frac{11\pi}{36}$$

$$n = 2 \Rightarrow x = 2\pi + \frac{\pi}{12} = \frac{25\pi}{12}$$

$$n = 3 \Rightarrow 3x = 3\pi - \frac{\pi}{12}$$

$$x = 3\pi - \frac{\pi}{12}$$

$$x = \frac{35\pi}{36}$$

5. Ans (B,C)

$$T_n = \frac{1}{n} \sum_{r=0}^{n-1} f\left(\frac{r}{n}\right)$$

$$\text{where } f(x) = x^4 + 2x^3 + x^2 + 3,$$

$$f'(x) = 4x^3 + 6x^2 + 2x > 0 \quad \forall x > 0$$

$$T_n = \frac{1}{n} \left(f(0) + f\left(\frac{1}{n}\right) + \dots + f\left(\frac{n-1}{n}\right) \right)$$

$$T_n < \int_0^1 (x^4 + 2x^3 + x^2 + 3) dx = \left[\frac{x^5}{5} + \frac{2x^4}{4} + \frac{x^3}{3} + 3x \right]_0^1$$

$$T_n < \frac{1}{5} + \frac{2}{4} + \frac{1}{3} + 3$$

$$T_n < \frac{121}{30}$$

$$S_n = \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right) = \frac{1}{n} \left\{ f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f\left(\frac{n}{n}\right) \right\}$$

$$> \int_0^1 (x^4 + 2x^3 + x^2 + 3) dx$$

$$S_n > \frac{121}{30}$$

6. Ans (A,D)

$$f'(x) = 2 \ln x \left(\frac{1}{x} \right) + \frac{2}{x} = \frac{2}{x} (\ln x + 1)$$

The tangent drawn from origin whose slope of

$$\text{tangent} = \frac{f(x)}{x}$$

$$\Rightarrow \frac{2}{x} (\ln x + 1) = \frac{(\ln x)^2 + 2 \ln x}{x}$$

$$\Rightarrow (\ln x)^2 = 2 \Rightarrow x = e^{\sqrt{2}}, e^{-\sqrt{2}}$$

$$5x \cdot f'(x) - x \cdot \ln 10 = 10$$

$$\Rightarrow 5x \cdot \frac{2}{x} \cdot (\ln x + 1) - x \cdot \ln 10 = 10$$

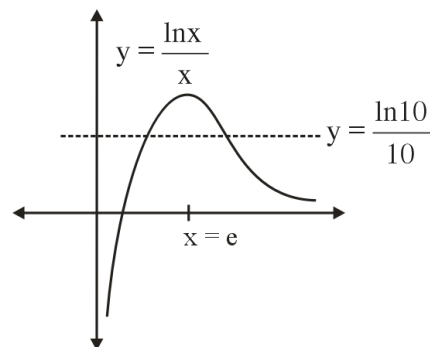
$$\Rightarrow 10 \cdot \ln x + 10 - x \cdot \ln 10 = 10$$

$$\Rightarrow \frac{\ln x}{x} = \frac{\ln 10}{10}$$

$$\text{Let } y = \frac{\ln x}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 - \ln x}{x^2}$$

$$\Rightarrow y_{\max} \text{ at } x = e$$



Hence equation has 2 solutions.

7. Ans (B,C)

$$\vec{b} \times \vec{d} = 0$$

$$\vec{c} \cdot \vec{d} = 0$$

Take cross with $\vec{d}$ in equation $\vec{a} = \vec{b} + \vec{c}$

$$\Rightarrow \vec{a} \times \vec{d} = 0 + \vec{c} \times \vec{d}$$

$$\Rightarrow \vec{d} \times (\vec{a} \times \vec{d}) = \vec{d} \times (\vec{c} \times \vec{d})$$

$$= (\vec{d} \cdot \vec{d}) \vec{c} - (\vec{d} \cdot \vec{c}) \vec{d}$$

$$= |\vec{d}|^2 \vec{c} - 0$$

$$\Rightarrow \frac{\vec{d} \times (\vec{a} \times \vec{d})}{|\vec{d}|^2} = \vec{c}$$

$$\Rightarrow \frac{|\vec{d} \times (\vec{a} \times \vec{d})|}{|\vec{d}|^2} = |\vec{c}| = 2$$

As we have $\vec{a} = \vec{b} + \vec{c}$

$$\Rightarrow \vec{a} \cdot \vec{d} = \vec{b} \cdot \vec{d} + \vec{c} \cdot \vec{d}$$

$$\Rightarrow \vec{a} \cdot \vec{d} = \vec{b} \cdot \vec{d} \quad \dots (1)$$

Also as we know $\vec{b} \times \vec{d} = 0$

take cross with $\vec{a}$ both sides

$$\vec{a} \times (\vec{b} \times \vec{d}) = 0$$

$$(\vec{a} \cdot \vec{d}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{d} = 0$$

$$(\vec{b} \cdot \vec{d}) \vec{b} = (\vec{a} \cdot \vec{b}) \vec{d} \quad (\text{from (1)})$$

8. Ans (B,C)

$$x_{i+2} = \frac{5}{x_{i+1}} + x_i$$

$$x_{i+2} \cdot x_{i+1} = 5 + x_i \cdot x_{i+1}$$

$$x_{i+2} \cdot x_{i+1} - x_i \cdot x_{i+1} = 5$$

$$\sum_{i=1}^{149} (x_{i+2} \cdot x_{i+1} - x_i \cdot x_{i+1}) = 745$$

$$x_{150} \cdot x_{151} - x_1 x_2 = 745$$

$$x_{150} \cdot x_{151} = 745 + 10 = 755$$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (D)

$$(a) A = \{1, 2, 3, 4, 5\}, B = \{1, 2, 3, 4\}$$

No. of onto function

$$= 4^5 - {}^4C_1 3^5 + {}^4C_2 2^5 - {}^4C_3 1^5$$

$$= 1024 - 972 + 192 - 4$$

$$= 240$$

$$(b) f(i) \leq f(j) \quad \forall i < j$$

Taking identical items in set A are distributed in set B

$$x_1 + x_2 + x_3 + x_4 = 5 \quad \text{each } x_i \geq 0, x_i \in \mathbb{I}, i = 1, 2, 3, 4.$$

$$\text{No. of solution} = {}^{5+4-1}C_{4-1} = {}^8C_3 = \frac{8 \times 7 \times 6}{6} = 56$$

10. Ans (C)

(c) The elements of the set $\{1, 2, 3, 4, 5\}$ are distributed in $\{1, 2, 3, 4\}$ where as $\{1, 2\}$ must be present in range of f

$$\text{so the no. of function} = 4^5 - {}^2C_1 3^5 + {}^2C_2 2^5$$

$$= 1024 - 486 + 32 = 570$$

(d) The number of function, where $f(1) = 4$

$$f(i) \geq f(j), \forall i, j \in \{2, 3, 4, 5\} = {}^{4+4-1}C_{4-1} = {}^7C_3 = 35$$

11. Ans (B)

$$(a) \text{Prob} = 1 - P(\text{all numbers are } \geq 10)$$

$$1 - \frac{{}^{16}C_4}{{}^{25}C_4} = \frac{1083}{1265}$$

$$(c) P = P(1 \text{ even, 3 odd}) + p(3 \text{ even, 1 odd})$$

$$= \frac{{}^{12}C_1 {}^{13}C_3 + {}^{13}C_1 {}^{12}C_3}{{}^{25}C_4} = \frac{286}{575}$$

12. Ans (C)

$$(b) \frac{(1)^9 {}^9C_1 \cdot {}^{15}C_2}{{}^{25}C_4} = \frac{189}{2530}$$

$$(d) P = 1 - P(\text{abcd is odd})$$

$$= 1 - \frac{{}^{13}C_4}{{}^{25}C_4} = \frac{217}{230}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.60)

$$\text{Given, } AA' = 4I \Rightarrow |A| = \pm 8$$

$$\text{So, } A' = 4A^{-1} = \frac{4 \text{adj}(A)}{|A|}$$

$$\begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} = \frac{4}{|A|} \begin{bmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{bmatrix}$$

$$\text{Now, } a_{ij} = \frac{4}{|A|} c_{ij}, \forall i, j$$

$$\Rightarrow -\frac{1}{2} c_{ij} = \frac{4}{|A|} c_{ij} \quad (\because 2a_{ij} + c_{ij} = 0)$$

$$\Rightarrow |A| = -8$$

Now,

$$\begin{aligned} |A + 4I| &= |A + AA'| = |A| |I + A'| = -8 |(I + A)'| \\ &= -8 |I + A| \end{aligned}$$

$$\Rightarrow |A + 4I| + 8 |A + I| = 0$$

$$\text{So, an comparing, we get } \lambda = \frac{8}{5} = 1.60$$

2. **Ans (2.82)**

Let

$$a^2 = \tan \alpha$$

$$b^2 = \tan \beta$$

$$c^2 = \tan \gamma, \text{ where } \alpha, \beta, \gamma \in \left(0, \frac{\pi}{2}\right)$$

$$\frac{1}{1+a^4} + \frac{1}{1+b^4} + \frac{1}{1+c^4} = \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\Rightarrow \sin^2 \alpha = \cos^2 \beta + \cos^2 \gamma$$

$$\text{Now, } \frac{\cos^2 \beta + \cos^2 \gamma}{2} \geq \sqrt{\cos^2 \beta \cdot \cos^2 \gamma}$$

$$\cos^2 \beta + \cos^2 \gamma \geq 2 \cos \beta \cdot \cos \gamma$$

$$\Rightarrow \sin^2 \alpha \geq 2 \cos \beta \cdot \cos \gamma \quad \dots(i)$$

$$\text{Similarly } \sin^2 \beta \geq 2 \cos \alpha \cdot \cos \gamma \quad \dots(ii)$$

$$\sin^2 \gamma \geq 2 \cos \alpha \cdot \cos \beta \quad \dots(iii)$$

Multiply equation (i), (ii) and (iii).

$$\sin^2 \alpha \cdot \sin^2 \beta \cdot \sin^2 \gamma \geq 8 \cos^2 \alpha \cdot \cos^2 \beta \cdot \cos^2 \gamma$$

$$\tan^2 \alpha \cdot \tan^2 \beta \cdot \tan^2 \gamma \geq 8$$

$$\tan \alpha \cdot \tan \beta \cdot \tan \gamma \geq 2\sqrt{2}$$

$$a^2 b^2 c^2 \geq 2\sqrt{2}$$

3. **Ans (15.00)**

Let $f(x)$ be a polynomial with integral coefficient.

$$f(\alpha) = 0 \quad \alpha \in I$$

$$f(x) = g(x)(x - \alpha)$$

$\Rightarrow g(x)$ be a polynomial has integral coefficient.

$$f(10) = 5$$

$$5 = g(10) \cdot (10 - \alpha)$$

$$\Rightarrow 10 - \alpha \text{ can be } 1, -1, 5, -5.$$

$$\Rightarrow \alpha = 9, 11, 5, 15 \quad \dots (1)$$

$$\text{and } f(2) = 13$$

$$2 - \alpha = 1, -1, 13, -13.$$

$$\alpha = 1, 3, -11, 15. \quad \dots (2)$$

So, common value of α which satisfy both equation (1) & (2) is

$$\alpha = 15 \text{ is common.}$$

4. **Ans (75.00)**

$$ab^2c^3 = 1024 \times 2^{71} = 2^{81}$$

$$\log_2 a + 2\log_2 b + 3\log_2 c = 81$$

$$\text{Let } \log_2 a = x \quad 2\log_2 b = y \quad 3\log_2 c = z$$

$$x + y + z = 81$$

Again

$$\log_2 a \times 2\log_2 b + \log_2 a \times 3\log_2 c + 2\log_2 b \times 3\log_2 c = 468$$

$$xy + xz + yz = 468$$

$$x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + yz + zx)$$

$$= (81)^2 - 2 \times 468$$

$$= 6561 - 936 = 5625$$

$$\sqrt{x^2 + y^2 + z^2} = 75$$

5. **Ans (2.25)**

Divide given diff. equation by $xy(y^2 - x^2)$.

$$\frac{y(y^2 - 2x^2)}{xy(y^2 - x^2)} dx + \frac{x(2y^2 - x^2)}{xy(y^2 - x^2)} dy = 0$$

$$\Rightarrow \frac{y^2 - x^2 - x^2}{x(y^2 - x^2)} dx + \frac{y^2 + y^2 - x^2}{y(y^2 - x^2)} dy = 0$$

$$\Rightarrow \frac{dx}{x} - \frac{x \cdot dx}{y^2 - x^2} + \frac{y dy}{y^2 - x^2} + \frac{dy}{y} = 0$$

$$\Rightarrow \frac{dx}{x} + \frac{dy}{y} + \frac{y dy - x dx}{y^2 - x^2} = 0$$

Integrating on both side

$$\Rightarrow \ln x + \ln y + \frac{1}{2} \ln(y^2 - x^2) = k$$

$$\Rightarrow x^2 \cdot y^2 (y^2 - x^2) = c$$

where c is const.

$$\Rightarrow f(x) = x^2, \quad g(y) = y^2$$

$$\Rightarrow \frac{g(3)}{f(2)} = \frac{9}{4} = 2.25$$

6. Ans (23.55)

As plane pass through a common line, their normals will be coplanar.

$$\begin{vmatrix} -1 & \sin \alpha & \sin \beta \\ \sin \alpha & -1 & \sin \theta \\ \sin \beta & \sin \theta & -1 \end{vmatrix} = 0$$

$\Rightarrow$

$$\sin^2 \theta + \sin^2 \alpha + \sin^2 \beta + 2 \sin \alpha \cdot \sin \beta \cdot \sin \theta = 1$$

...(1)

Now $\vec{P}_1 \times \vec{P}_2$ is $\parallel$ to given line

$$\vec{P}_1 \times \vec{P}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & \sin \alpha & \sin \beta \\ \sin \alpha & -1 & \sin \theta \end{vmatrix}$$

$$= \hat{i} (\sin \alpha \cdot \sin \theta + \sin \beta) + \hat{j} (\sin \theta + \sin \alpha \cdot \sin \beta) + \hat{k} (1 - \sin^2 \alpha)$$

$\Rightarrow$

$$\frac{\sin \alpha \cdot \sin \theta + \sin \beta}{\cos \theta} = \frac{\sin \theta + \sin \alpha \cdot \sin \beta}{\cos \beta} = \frac{1 - \sin^2 \alpha}{\cos \alpha} = \cos \alpha$$

...(2)

$\Rightarrow$

$$\sin \theta = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta = \cos(\alpha + \beta)$$

Also,

$$\sin \beta = \cos \alpha \cdot \cos \theta - \sin \alpha \cdot \sin \theta = \cos(\alpha + \theta)$$

$$\Rightarrow \alpha + \beta + \theta = 2n\pi + \frac{\pi}{2}$$

$$\text{Possible values} = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}$$